SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA5116
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

Annexure A

Code of Conduct:

I _____________varsha.s/192511358_______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Analytical Problem Solving

1. **RSA Key Generation and Security Analysis**Given: p = 17, q = 19

Compute n and ϕ(n)
Choose e = 5 and calculate the private key d
Encrypt the message M = 7
Decrypt the ciphertext and verify correctness
Analyze the impact of choosing a small prime number on RSA security

2. ElGamal Encryption Analysis
Given:
p = 23, g = 5, private key x = 7, random k = 3, message M = 10

Compute the public key
Perform encryption to generate ciphertext pair (C1, C2)
Decrypt the ciphertext to retrieve the original message
Analyze how randomness (k) improves security in ElGamal

3. RSA Digital Signature Verification
Given:
p = 11, q = 17, e = 7, message M = 8

Compute n and $\phi(n)$
Find private key d
Generate the digital signature for the message
Verify the signature using the public key
Analyze how digital signatures ensure non-repudiation

4. Diffie-Hellman with Attack Resistance Evaluation
Given:
p = 29, g = 2 User A
private key = 5
User B private key = 12

Compute public keys for both users
Compute the shared secret key
Analyze what happens if an attacker intercepts and replaces public keys
Suggest a cryptographic solution to ensure authenticity

5. Hybrid Encryption System Analysis
Given:
A system uses RSA for key exchange and AES for data encryption
RSA encryption time = 5 ms per key exchange
AES encryption time = 0.5 ms per data block
Total data blocks = 500

Calculate total time taken for encryption
Compare with using only RSA for encrypting all data
Analyze why hybrid encryption is more efficient Justify
its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. RSA Key Generation and Security Analysis
Given:

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-
- Message

Step 1: Compute and
Step 2: Calculate Private Key
Find such that
Using the Extended Euclidean Algorithm,
Public Key = (5,323)
Private Key = (173,323)

Step 3: Encrypt the Message
Encryption Formula:
Ciphertext = 11

Step 4: Decrypt the Ciphertext
Decryption Formula:
The decrypted message is equal to the original message.

Security Analysis

- RSA security depends on the difficulty of factoring large numbers.
- Small prime numbers make factorization easy.
- Attackers can quickly determine the private key.
- Therefore, large primes (2048-bit or higher) should always be used.

Result

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- Private key
- Ciphertext = **11**
- Decrypted message = **7**

2. ElGamal Encryption Analysis
Given

- Prime number
- Generator
- Private key
- Random number
- Message

Step 1: Compute Public Key
Formula
Public Key = (23,5,17)

Step 2: Encryption
Compute
Next,
Ciphertext = (10,2)

Step 3: Decryption
Compute
Inverse of 14 modulo 23 = 5
Recovered message = **10**

Security Analysis

- Random value is different for every encryption.
- The same plaintext produces different ciphertexts.
- This prevents replay attacks and improves security.
- Even if an attacker knows the public key, the message cannot be easily recovered.

Result

- Public key = **(23,5,17)**
- Ciphertext = **(10,2)**
- Decrypted message = **10**

3. RSA Digital Signature Verification
Given

-
-
- Message

Step 1: Compute and

Step 2: Find Private Key
Find
Private Key = **(23,187)**

Step 3: Generate Digital Signature
Formula
Digital Signature = **162**

Step 4: Verify Signature
Formula
Recovered message equals the original message.
Therefore,
Signature is Valid.

Analysis
Digital signatures provide

- Authentication
- Data Integrity
- Non-repudiation

Only the sender possesses the private key, so the sender cannot deny sending the message.

Result

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- Signature = **162**
- Verification Successful

4. Diffie–Hellman Key Exchange
Given
- Prime number
- Generator
- User A private key = 5
- User B private key = 12

Step 1: Compute Public Keys
User A
User B

Step 2: Compute Shared Secret
User A
User B
Both obtain the same shared secret.

Attack Analysis
If an attacker intercepts and replaces the exchanged public keys, a **Man-in-the-Middle (MITM)** attack occurs. The attacker establishes separate secret keys with both users and can read or modify communication.

Cryptographic Solution
To ensure authenticity:
- Digital Certificates
- RSA Digital Signatures
- Public Key Infrastructure (PKI)

These methods verify public keys before key exchange.

Result
- Public Key of A = **3**
- Public Key of B = **7**
- Shared Secret = **16**

5. Hybrid Encryption System Analysis
Given
- RSA key exchange = **5 ms**
- AES encryption = **0.5 ms/block**
- Number of blocks = **500**

Step 1: AES Encryption Time

Step 2: Total Encryption Time
Total Hybrid Encryption Time = **255 ms**

Step 3: RSA-only Encryption
RSA-only Encryption Time = **2500 ms**

Comparison

Method	Time
Hybrid Encryption	**255 ms**
RSA Only	**2500 ms**

Hybrid encryption is approximately **10 times faster** than using RSA alone.

Analysis

- RSA is computationally expensive and is suitable only for key exchange.
- AES is a fast symmetric encryption algorithm used for encrypting bulk data.
- Combining RSA and AES provides both security and efficiency.
- Therefore, hybrid encryption is widely used in HTTPS, VPNs, secure email, cloud storage, and online banking.

Result

- AES Time = **250 ms**
- RSA Time = **5 ms**
- Total Hybrid Time = **255 ms**
- RSA-only Time = **2500 ms**
- **Hybrid encryption is faster, secure, and efficient.**